

TEXTBOOK TITLE

An example textbook template with margin notes

Contents

I	The first chapter	I
1.1	Lorem ipsum	I
1.2	Mathematical symbols $HH \rightarrow b\bar{b}\tau^+\tau^-$	3
1.3	Examples of unit typesetting	3
1.4	Examples of unit typesetting using electron-volt units with better kerning	3
1.5	Examples of shorthand commands	4

Chapter I

The first chapter

I.I Lorem ipsum

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nam magna ex, dictum id convallis vel, tincidunt non lectus.¹ Nulla porttitor turpis in felis malesuada, ut semper risus hendrerit. Donec condimentum nulla odio, id aliquet lectus aliquet a.

This is an example margin note.

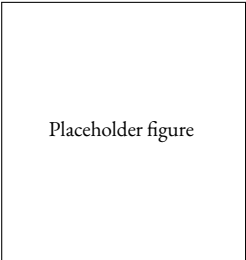
Integer cursus dolor a felis ornare lobortis. Duis varius dolor felis, in elementum nisl vestibulum vel. In sollicitudin justo nisl, sit amet dignissim libero porta hendrerit.[1] Morbi molestie tincidunt justo a lobortis.[2]

This is an example margin note shifted vertically.

Etiam ac lectus ac nisl rutrum egestas. Nam ligula nisi, ultricies aliquet ligula ac, sagittis sodales dolor. Sed volutpat lacus at urna convallis, a feugiat augue tincidunt. Nam ex ipsum, placerat ut sem vel, condimentum cursus eros. Donec leo felis, auctor sit amet arcu ut, vehicula congue dolor. Duis augue ex, varius sit amet elit id, blandit facilisis orci. Aliquam euismod eget mi in ullamcorper. Donec hendrerit imperdiet lobortis.

Duis eu neque non felis aliquet posuere. Quisque mollis eros vitae iaculis rhoncus. Nunc at arcu sapien. Aenean aliquam, metus consectetur iaculis elementum, tortor tellus placerat nisi, a condimentum nulla enim et nisi. Maecenas semper volutpat felis tristique bibendum.

Nunc efficitur, nulla et scelerisque euismod, nunc magna pellentesque ligula, quis laoreet justo lorem eu augue. Vestibulum vestibulum



Margin placeholder figure.

¹ This is an example footnote.
[1] L. Li Tianjun, W. Xia, W. You-kai and Z. Shou-hua, *Distinguishing the Color Octet Axial-Vector-like Particle for Top Quark Asymmetry via Color Flow Method at the LHC* (June 2013), arXiv:1306.3586
[2] W. S. McCulloch and W. Pitts, *A logical calculus of the ideas immanent in nervous activity*, The Bulletin of Mathematical Biophysics, 5 (4), 115–133 (1943)

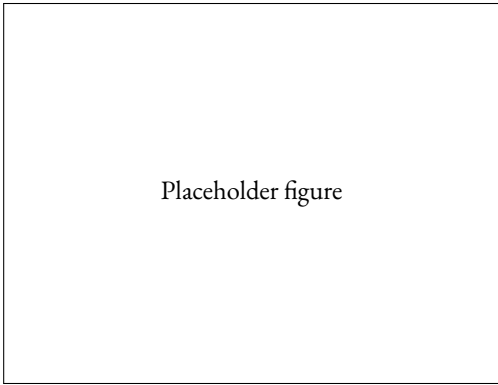


Figure 1.1: Placeholder figure.

lum, ipsum vitae ultrices commodo, lorem diam semper nisi, non finibus ante justo sed libero. Phasellus molestie maximus nulla id sodales. Integer pulvinar elit ut justo rhoncus cursus. Nullam pharetra dolor vitae mi vulputate, nec congue dui iaculis. Curabitur vehicula molestie mauris, imperdiet semper metus bibendum a. Aliquam vitae est lorem. Cras mollis vel mauris eget interdum. Cras gravida, nibh nec dapibus bibendum, nibh velit imperdiet ligula, in ultrices magna mauris nec orci.

Maecenas faucibus dictum libero. Nam pellentesque eros mi, vitae malesuada ex semper a. Suspendisse vitae ultrices diam, sed feugiat mauris. Donec malesuada commodo turpis sit amet consectetur. Mauris posuere id mauris eu accumsan. Cras eu tincidunt magna. Aliquam erat volutpat. Nulla volutpat elit quis lorem vehicula convallis. Sed eu massa sollicitudin dolor eleifend faucibus id non odio. Aenean blandit elit mauris, placerat posuere dui laoreet eu. Nullam feugiat sollicitudin lectus. Curabitur ac tortor ut turpis molestie aliquet. Sed eget varius arcu, placerat imperdiet lorem. Fusce commodo pretium eros, rhoncus egestas velit porttitor non. Phasellus ornare pretium iaculis. See Figure 1.2.



Figure 1.2: HH logo.

1.2 Mathematical symbols $HH \rightarrow b\bar{b}\tau^+\tau^-$

The di-Higgs final state $HH \rightarrow b\bar{b}\tau^+\tau^-$ features heavy-flavour jets and leptonic or hadronic τ decays.

Equations without references:

$$E = mc^2.$$

Equations with references:

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}. \quad (1.1)$$

1.3 Examples of unit typesetting

- Speed of light: $c = 2.997\,924\,58 \times 10^8$ m/s
- Beam energy: 6.5 TeV per beam
- Tracker pitch: 50 μm to 75 μm
- Temperature variation: ± 0.3 K
- Luminosity: 2.0/fb
- 12.3 kg
- 3.45×10^8

1.4 Examples of unit typesetting using electron-volt units with better kerning

- Usage in normal text: the beam energy is 6.5 TeV per beam.
- Usage in mathematics environment: $E_{\text{beam}} = 6.5$ TeV.
- Usage with `sunitx`: The beam energy is 6.5 TeV per beam.
- The PETRA accelerator beams at DESY were of 24 GeV/ c each.

1.5 Examples of shorthand commands

- Jets are reconstructed using the anti- k_t clustering algorithm.
- The k_t algorithm clusters softer constituents first.
- A Higgs boson can decay to a $b\bar{b}$ pair.
- A b -hadron contains at least one b -quark.
- Secondary vertices can reveal long-lived b -hadrons.
- The leading b -jet has the highest transverse momentum.
- Events with two b -jets are retained for further analysis.
- A b -quark hadronises before it can be observed directly.
- Two b -quarks are produced in this Higgs-boson decay.
- The b -tagging efficiency is measured in collision data.
- Each selected jet must satisfy the b -tag requirement.
- The event contains exactly two b -tags.
- A c -hadron contains at least one c -quark.
- Misidentified c -hadrons form an important background.
- CPT symmetry relates particles to their antiparticles.
- The η - ϕ separation helps distinguish nearby objects.
- Jets, leptons, photons *etc.* may be reconstructed in the detector.
- Charged leptons, *e.g.* electrons, leave tracks in the inner detector.
- The candidate is neutral, *i.e.* it carries no electric charge.
- A 5σ excess is conventionally associated with a discovery.
- The $H \rightarrow b\bar{b}$ channel has the largest Higgs-boson branching fraction.
- The H_T variable is the scalar sum of transverse momenta.
- The H_T^{had} variable includes only the hadronic contribution.
- A large- R jet can contain all the decay products of a boosted particle.
- Large- R jets are useful for reconstructing boosted heavy particles.
- Large $\cancel{E}_T$ can indicate invisible particles in the final state.

- The selection may be written as $\cancel{E}_T > 200 \text{ GeV}$.
- The M_{T2} endpoint can constrain the mass scale of a process.
- The leading-lepton p_T spectrum is compared with the prediction.
- The p -value quantifies the compatibility of the data with a hypothesis.
- Tracks curve in the r - ϕ plane when a magnetic field is present.
- The weak interaction is based on the $\text{SU}(2)_L$ gauge group.
- The S_T requirement suppresses low-energy background events.
- A 3σ deviation is evidence, but not normally a discovery.
- Inclusive $t\bar{t}$ production is a major background at the LHC.
- The $t\bar{t} + b\bar{b}$ process contains an additional b -quark pair.
- The $t\bar{t} + \text{jets}$ sample includes events with additional jets.
- The $t\bar{t}b\bar{b}$ final state contains two t -quarks and two b -quarks.
- The $t\bar{t} + b\bar{b}$ notation highlights b -jets produced with a t -quark pair.
- The $t\bar{t}H$ process directly probes the t -quark Yukawa coupling.
- The $t\bar{t}H$ ($b\bar{b}$) channel combines associated production with a b -quark decay.
- The $t\bar{t}$ ($H \rightarrow b\bar{b}$) notation makes the Higgs-boson decay explicit.
- The t -quark is the heaviest elementary particle yet observed.
- At the LHC, t -quarks are produced both singly and in pairs.
- A $W^\pm$ -boson mediates charged-current weak interactions.
- Pairs of $W^\pm$ -bosons provide a sensitive test of electroweak theory.
- The symbol $W^\pm$ includes both charged states of the weak boson.
- A Z^0 -boson mediates neutral-current weak interactions.
- The symbol Z^0 denotes the neutral weak boson.

Index

energy, 3

light

 speed of, 3

temperature, 3

Bibliography

- [1] L. Li Tianjun, W. Xia, W. You-kai and Z. Shou-hua, *Distinguishing the Color Octet Axial-Vector-like Particle for Top Quark Asymmetry via Color Flow Method at the LHC* (June 2013), arXiv:1306.3586 ¹
- [2] W. S. McCulloch and W. Pitts, *A logical calculus of the ideas immanent in nervous activity*, The Bulletin of Mathematical Biophysics, 5 (4), 115–133 (1943) ¹